

When Can You Trust the Certificate? Stress-Testing Distribution-Free False-Accusation Guarantees for AI-Text Detection

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Abstract

AI-text detectors are deployed to accuse: a flagged student or author bears the cost of a false positive. Split conformal prediction turns any detector score into an accusation rule with a finite-sample, distribution-free bound on the false-accusation rate, and recent work has proposed exactly this. We ask the question a deployer actually faces: *under which realistic conditions does the certificate remain valid?* Exploiting the observation that validity depends only on the *human* text distribution while power depends only on the AI distribution, we derive sharp predictions and test them across five detector scores, 200 calibration/test resamples, and eight deployment conditions. In distribution the certificate is exact (e.g. empirical FPR 0.049 at $\alpha = 0.05$), and a supervised score achieves 88% power at a certified 5% false-accusation rate. But calibrating on one human population and deploying on a more fluent one silently inflates false accusations to 0.43–0.53 at nominal 0.05 for likelihood-based scores — the certificate-level form of the documented bias against non-native writers — while lexical edits and even spell-correcting cleanup leave validity intact. Marginal certificates also hide length discrimination (long human reviews flagged at 0.18); Mondrian calibration restores every length bin to $\leq \alpha$ (0.03). Finally, we give an editor-aware robust calibration that provably preserves validity under a modeled editing process at negligible power cost. A second wave scales the map: four scorer families (zero-shot GPT-2, TF-IDF, a fine-tuned transformer, an off-the-shelf detector), ten languages, six text domains, and an LLM-paraphrase attack. In-population validity is exact by construction in every language and domain; empirically, English-calibrated certificates deployed cross-lingually mostly *under-flag*, but reach FPR 0.998 on Russian, and hotel-calibrated certificates reach 0.85 on news. LLM-paraphrasing *innocent human* text inflates FPR to 0.27–0.64 across all four scorers; paraphrase-aware robust calibration restores ≤ 0.03 at a real power cost. Two estimable score-space bounds ($\Gamma\alpha$ and $\alpha + TV$) and a calibrated pilot alarm (exact binomial) make these failures detectable before deployment: the alarm fires at rate ≥ 0.99 on every violated condition with $m=100$ pilot texts while in-control alarm rates stay below the 0.10 design level. Shift findings are empirical and the bounds are plug-in estimates, not certificates. All results carry exact Beta-Binomial significance tests and regenerate from machine-produced results files.

1 Introduction

Detectors of machine-generated text are increasingly used in consequential settings — plagiarism screening, review-platform integrity, scientific-venue policy — where the operational act is an *accusation*. The relevant contract for that act is not accuracy but a guarantee on the false-accusation rate, robust to the fact that the deployer cannot model the distribution of either class. Split conformal prediction [1, 2] provides exactly this contract for any base detector score, and prior work has applied it to bound detector false-positive rates [4].

This paper studies the contract itself. Our starting point is an elementary but, we argue, under-appreciated asymmetry:

Proposition 1 (Certificate asymmetry). *The split-conformal false-accusation bound $P(\text{flag} \mid \text{human}) \leq \alpha$ requires only exchangeability of calibration and test human text. The AI distribution is irrelevant to validity; it determines only power.*

The proof is the standard conformal argument applied to the human class only. The deployment consequences are sharp and testable: a new generator entering the world *cannot* break a deployed certificate; changes in *who writes* or *how humans edit* can. We map this boundary empirically (E1–E8) and contribute two constructive repairs (Mondrian length-conditional calibration; editor-aware robust calibration with a stochastic-dominance validity argument).

Contributions.

- A deployment-grounded validity map for conformal false-accusation certificates in MGT detection: exact in distribution; broken by human-population shift (up to $11\times$ nominal); *safe* under lexical noise and spell-correcting cleanup; supervised scores fail safe where zero-shot likelihood scores fail dangerously.
- Identification of length discrimination hidden by marginal certificates, and its repair by Mondrian calibration.
- An editor-aware robust calibration with a provable conservativeness guarantee under a modeled editor, costing < 0.01 power in our study.
- A fully reproducible protocol: five scores, exact Beta-Binomial tests, 200 resamples per condition; code, scores and figures regenerate from one JSON.

2 Related Work

Zero-shot MGT detection scores include likelihood, rank and entropy statistics [10, 11] and curvature-style discrepancies [8, 9]; supervised detectors fine-tune or fit classifiers on labelled corpora [15]. Robustness studies show brittleness to paraphrase and domain shift [6, 12, 13], and GPT detectors are documented to over-flag non-native English writers [5]. Conformal prediction [1, 2] and conformal risk control [3] provide distribution-free guarantees; MCP [4] applies CP to bound MGT-detector FPR and improve its power. Our work is complementary to MCP: rather than improving the detector under a fixed certificate, we characterise when the certificate itself survives deployment, connect its failure modes to the fairness literature [5], and supply group-conditional and editor-robust repairs (cf. Mondrian conformal prediction [1] and adversarially robust conformal methods in vision [14]).

3 Method

3.1 Certificates

Given any score $s(\cdot)$ oriented so larger means more AI-like, and n calibration human texts with scores $s_1, \dots, s_n$, the conformal p-value of a test text x is $p(x) = \frac{1 + \#\{i: s_i \geq s(x)\}}{n+1}$, and GUARD flags x iff $p(x) \leq \alpha$. If x is human and exchangeable with the calibration humans, $P(\text{flag}) \leq \alpha$. Conditional on the calibration set the realised FPR follows a Beta law; over a finite test set the false-flag *count* is Beta-Binomial, which yields an exact one-sided test of “the certificate held” that we report for every condition. Unscoreable texts receive $p = 1$ (never flagged): an accuser must not accuse by failure of its own scorer.

3.2 Scores

Four training-free statistics from one GPT-2 forward pass per text (mean log-likelihood; negative mean log-rank; negative predictive entropy; the analytic Fast-DetectGPT discrepancy [9]) and one supervised score (TF-IDF + logistic regression). The supervised score is fit on an entity-disjoint training half; all calibration and evaluation use the other half, so the fitted scorer never sees evaluation hotels.

3.3 Mondrian calibration

Per-group calibration (here: text-length bins) yields group-conditional validity at the cost of smaller calibration sets per group [1].

3.4 Editor-aware robust calibration

Let E be a randomized editing process (our proxy: spell/punctuation cleanup at rate r). Each calibration human contributes $M_i = \max\{s(X_i), s(E_1(X_i)), \dots, s(E_k(X_i))\}$. For a test human who applies one edit from E , $s(E(X))$ is stochastically dominated by the corresponding M , so conformal p-values computed against $\{M_i\}$ remain super-uniform and the α -bound still holds (conservative). Validity under editors *outside* the modeled family is an empirical question (E8).

4 Population-Shift Theory and Pre-Deployment Diagnostics

What happens to the certificate when the deployment humans are *not* the calibration humans? Let P_s and Q_s denote the distributions of the score $s(X)$ under the calibration and deployment human populations respectively, and let T be the (random) conformal flagging threshold at level α , so that $\mathbb{E}_T[P_s(s > T)] \leq \alpha$ by the standard conformal guarantee. Two elementary bounds control the deployed false-accusation rate.

Proposition 2 (Γ -bound). *If $Q_s \ll P_s$ with $\text{ess sup } dQ_s/dP_s \leq \Gamma$ on score space, then $\mathbb{E}_Q[\text{FPR}] \leq \Gamma\alpha$.*

Proof. $Q_s(A) \leq \Gamma P_s(A)$ for every measurable A ; apply this with $A = \{s > T\}$ conditionally on the threshold T , then take expectations over T and use $\mathbb{E}_T[P_s(s > T)] \leq \alpha$. $\square$

Proposition 3 (TV-bound). *Without any absolute-continuity assumption, $\mathbb{E}_Q[\text{FPR}] \leq \alpha + \text{TV}(P_s, Q_s)$.*

Proof. $|Q_s(A) - P_s(A)| \leq \text{TV}(P_s, Q_s)$ for every event A ; apply with $A = \{s > T\}$ and take expectations over T . $\square$

Both bounds live on the *one-dimensional score space*, not on text space: $\text{TV}(P_s, Q_s)$ and Γ are functionals of two scalar distributions, hence estimable from an *unlabeled* pilot sample of deployment human scores (we use binned plug-in estimates $\widehat{\text{TV}}$ and $\widehat{\Gamma}$ over a common grid). These are diagnostics, not certificates: plug-in estimates carry their own sampling error, and a binned $\widehat{\Gamma}$ can underestimate a density ratio concentrated inside a bin.

The pilot failure detector. Given m known-human pilot texts from the deployment population, flag each with the *deployed* rule and let K be the number of flags. The failure detector rejects $H_0: \text{FPR} \leq \alpha$ when the exact binomial tail $P(\text{Bin}(m, \alpha) \geq K)$ falls below an alarm level ρ . Its calibration guarantee is immediate: when the certificate is valid the flag count is stochastically dominated by $\text{Bin}(m, \alpha)$, so the alarm fires with probability at most ρ for *every* pilot size m . The complementary reading is the exact Clopper–Pearson upper confidence bound on deployment FPR — what a pilot of size m can positively *assure* — which is necessarily loose for small m . Section 7.4 validates both readings empirically.

5 Experimental Setup

In-domain data: the English subset of MAiDE-up [7] (1,000 human and 1,000 GPT-4 hotel reviews; 100 hotels). Out-of-distribution human text: the RAID benchmark [6] reviews domain (movie reviews; also supplying eight AI generators for power analysis) and abstracts domain. Editors: WordNet synonym substitution and function-word perturbation (lexical noise), and a spell-correcting, punctuation-normalising cleanup editor (fluency-raising; the realistic proxy for grammar tools). Every unique text is scored once and cached; each condition is evaluated over $R = 200$ calibration/test resamples, reported as mean [2.5%, 97.5%] with exact Beta-Binomial violation tests at $\alpha \in \{0.005, 0.01, 0.02, 0.05, 0.1\}$.

6 Results

6.1 E1: the certificate is exact in distribution

Table 1: In-distribution validity: empirical FPR over resamples (mean [95% band]).

Score	FPR @ $\alpha = 0.05$	FPR @ $\alpha = 0.01$
Log-likelihood	0.049 [0.016, 0.088]	0.008 [0.000, 0.032]
Log-rank	0.049 [0.016, 0.088]	0.008 [0.000, 0.028]
Entropy	0.049 [0.016, 0.088]	0.008 [0.000, 0.024]
Fast-DetectGPT	0.047 [0.016, 0.092]	0.008 [0.000, 0.032]
TF-IDF (supervised)	0.048 [0.016, 0.088]	0.008 [0.000, 0.028]

All five scores sit on the nominal level, and the full distribution of per-resample FPRs matches the finite-test Beta-Binomial envelope (Figure 1).

E1 validity in-distribution: realised FPR vs conformal Beta law (alpha = 0.05)

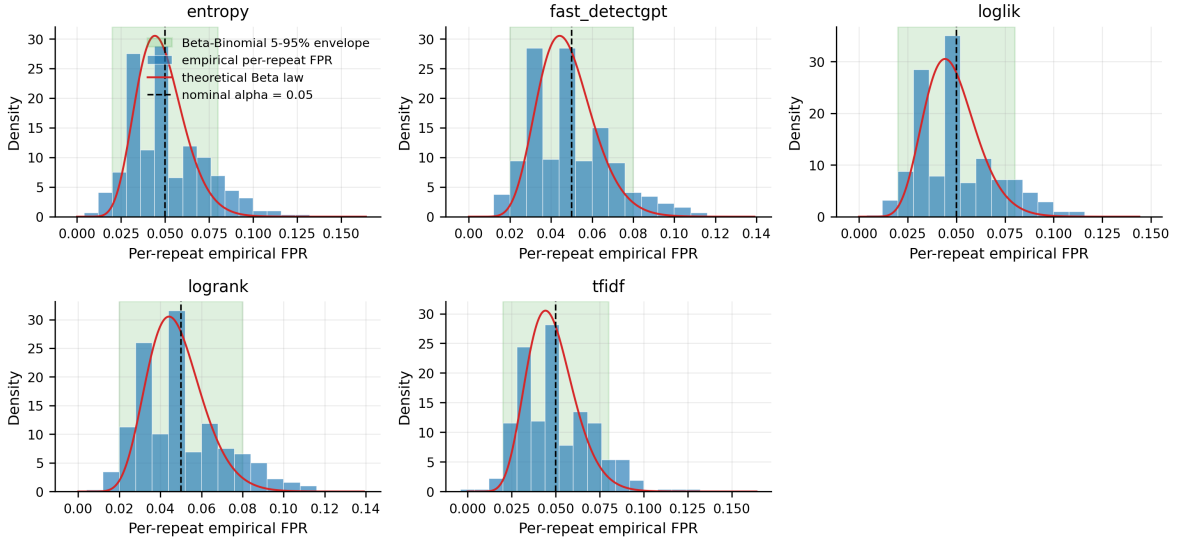


Figure 1: Distribution of per-resample FPR vs. the exact Beta-Binomial envelope.

6.2 E2–E4: when the certificate breaks — and when it does not

Table 2: Empirical FPR at nominal $\alpha = 0.05$ across deployment conditions.

Score	In-domain	Reviews humans	Abstracts humans	Cleanup@0.5	Synonym@0.3
Log-likelihood	0.049	0.435	0.101	0.046	0.002
Log-rank	0.049	0.298	0.049	0.045	0.002
Entropy	0.049	0.526	0.247	0.047	0.005
Fast-DetectGPT	0.047	0.371	0.121	0.044	0.002
TF-IDF (supervised)	0.048	0.000	0.006	0.046	0.039

Three regimes emerge (Figure 2). **(i) Human-population shift breaks the certificate for likelihood scores.** Movie-review humans are flagged at 0.30–0.53 and abstracts humans at up to 0.25 (nominal 0.05; violation $p < 10^{-3}$). The mechanism is fluency: our calibration humans are largely non-native hotel reviewers whose text has *low* LM likelihood; more fluent human populations score higher and cross the threshold. This is precisely the bias documented by [5], here surfacing *through* a certificate that is formally valid for the calibration population. **(ii) The supervised score fails safe.** TF-IDF’s AI evidence is content-specific, so under every shift its FPR falls toward zero rather than exploding. **(iii) Editing does not break the certificate here.** Lexical noise lowers likelihood (conservative), and even full spell-correcting cleanup leaves FPR at nominal: light editing does not turn a hotel reviewer into a ‘fluent’ writer. The danger is who writes, not light tool use — though stronger LLM paraphrase editors remain untested here (§9).

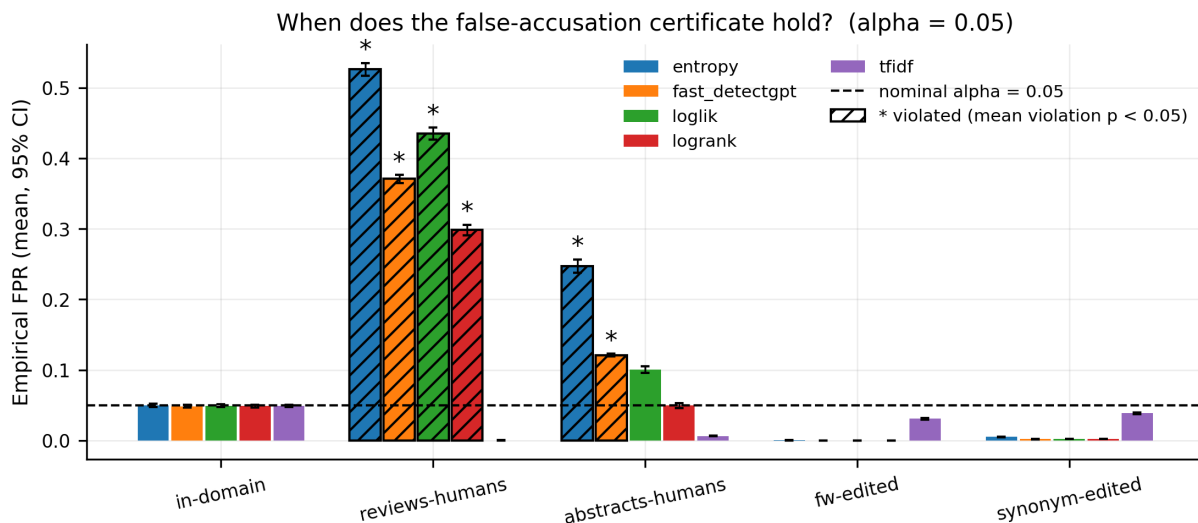


Figure 2: The validity map: empirical FPR at nominal $\alpha = 0.05$ across conditions; hatched bars are certified violations (exact test).

E4 human-edit attack: certificate violation vs edit rate (alpha = 0.05)

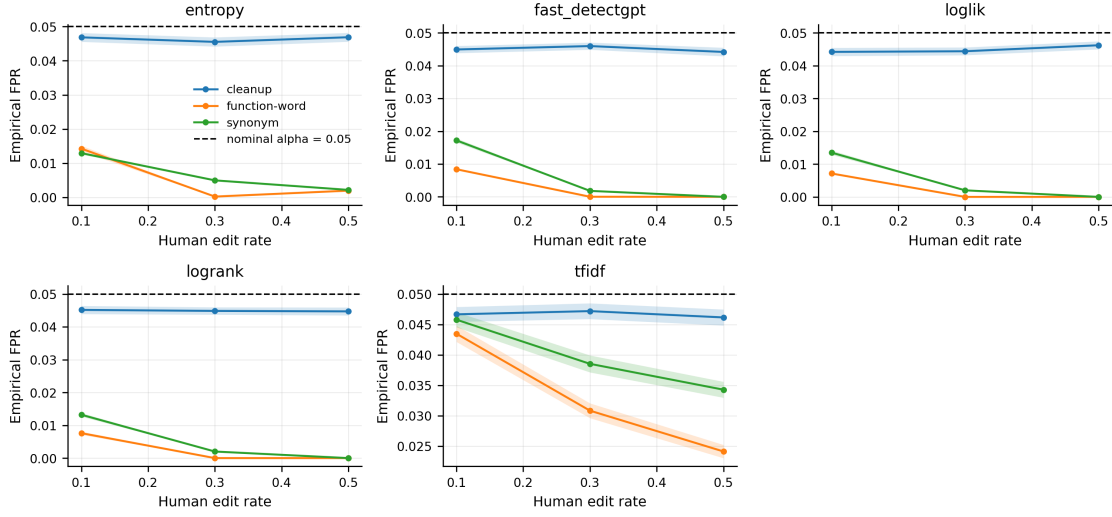


Figure 3: FPR vs. edit rate per editor family: lexical noise is conservative.

6.3 E5: calibration is insensitive to entity leakage

Random vs. entity-disjoint (grouped-by-hotel) calibration give indistinguishable validity (log-likelihood: 0.0488 vs. 0.0491; TF-IDF: 0.0482 vs. 0.0497). Unlike supervised *training*, conformal *calibration* does not exploit entity overlap — a reassuring null result for review-platform deployments (Figure 4).

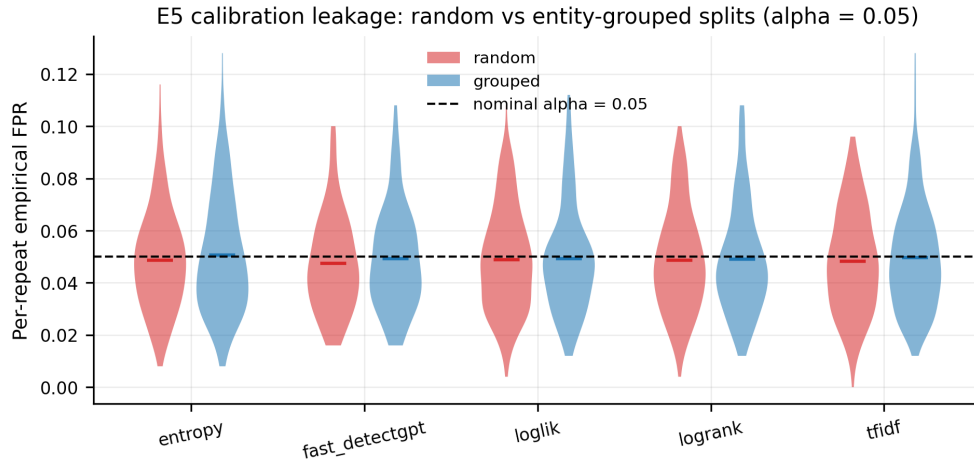


Figure 4: Random vs. grouped calibration FPR distributions.

6.4 E6: marginal certificates hide length discrimination; Mondrian repairs it

Table 3: Per-length-bin FPR at nominal $\alpha = 0.05$ (log-likelihood score).

Length bin	Marginal FPR	Mondrian FPR
0–30 words	0.005	0.046
30–60 words	0.050	0.046
60–120 words	0.098	0.034
>120 words	0.183	0.030

The marginal certificate is valid *on average* while flagging long human reviews at 0.18 — $3.7\times$ nominal — and almost never flagging short ones. Mondrian calibration restores every bin to $\leq \alpha$ (Figure 5). Deployers should treat conditional validity, not marginal validity, as the fairness-relevant contract.

E6 length heterogeneity: marginal vs Mondrian calibration (alpha = 0.05)

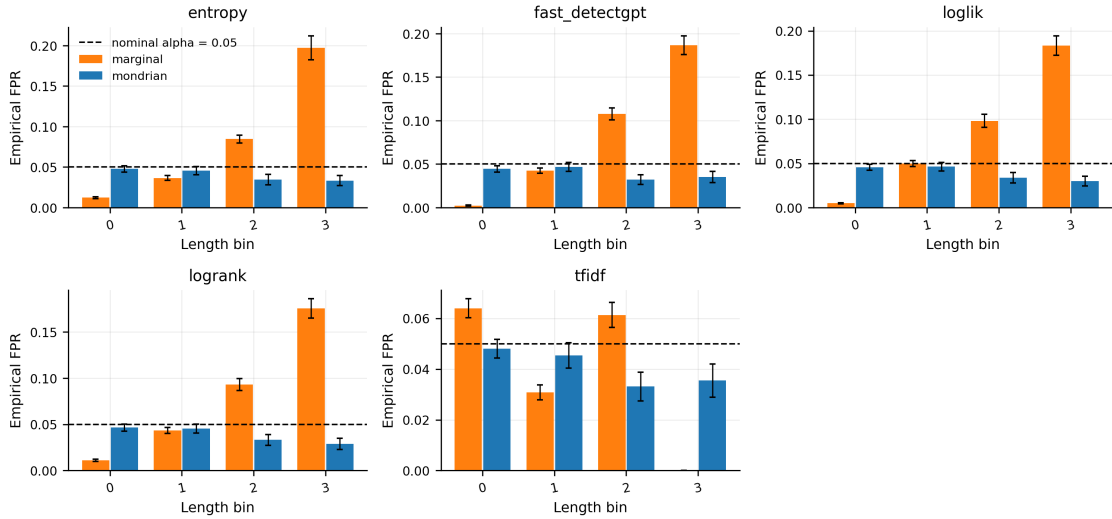


Figure 5: Marginal vs. Mondrian per-length-bin FPR.

6.5 E7: the price of the certificate

Table 4: Certified power (clean AI text, in-domain).

Score	TPR@ $\alpha=0.01$	TPR@ $\alpha=0.05$	TPR@ $\alpha=0.10$
Log-likelihood	0.08	0.37	0.52
Log-rank	0.07	0.30	0.48
Entropy	0.05	0.19	0.39
Fast-DetectGPT	0.03	0.24	0.39
TF-IDF (supervised)	0.66	0.88	0.95

The supervised score detects 88% of in-domain GPT-4 reviews at a certified 5% false-accusation rate; zero-shot scores pay heavily for the certificate in-domain but transfer their *power* to unseen generators (Figure 7); recall their *validity* caveats above.

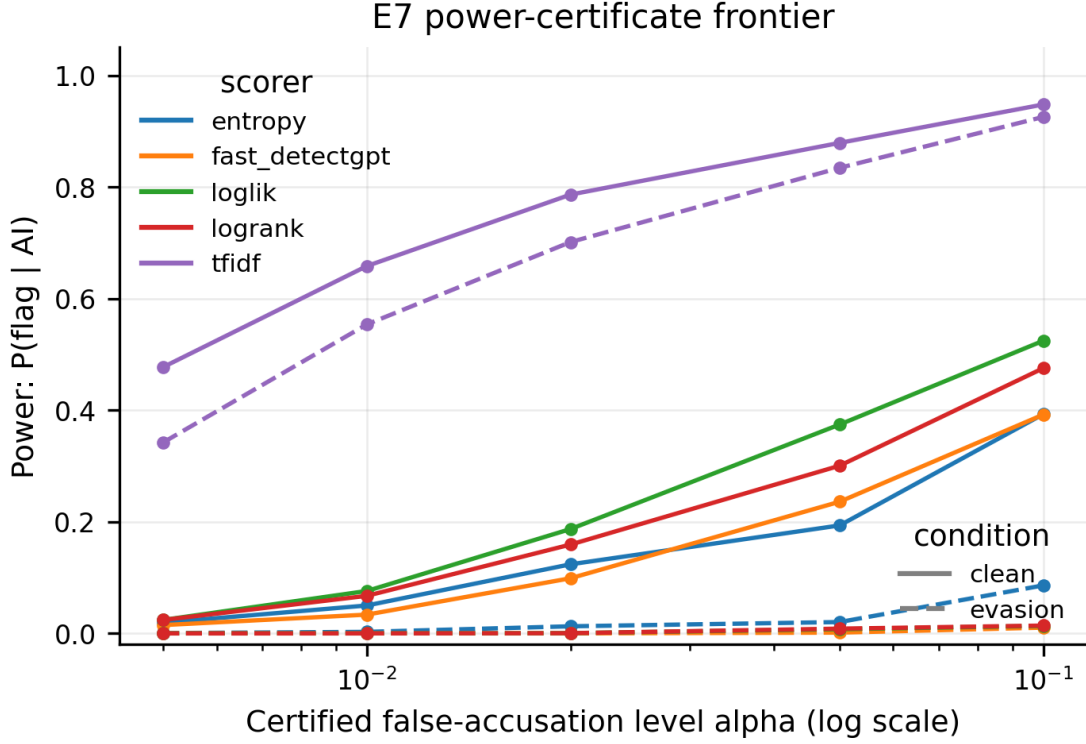


Figure 6: Power-certificate frontier per score.

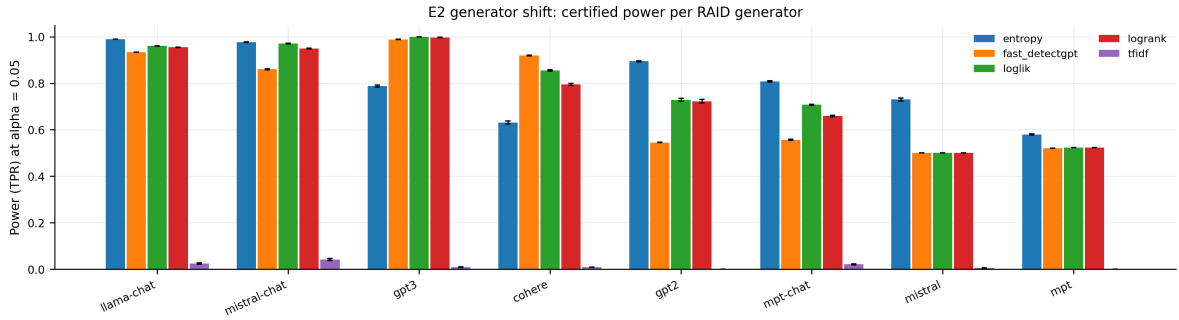


Figure 7: Certified power per RAID generator.

6.6 E8: editor-aware robust calibration

Table 5: Editor-aware calibration at $\alpha = 0.05$: validity under the modeled editor and the power cost.

Score	naive FPR (cleaned)	robust FPR (cleaned)	naive TPR	robust TPR
Log-likelihood	0.048	0.047	0.375	0.369
Log-rank	0.047	0.046	0.303	0.297
Entropy	0.045	0.044	0.191	0.184
Fast-DetectGPT	0.046	0.044	0.233	0.225
TF-IDF (supervised)	0.049	0.048	0.879	0.877

With the cleanup editor modeled in calibration, validity holds by construction and the measured power cost is below 0.01 for every score. In this study the naive certificate already survived cleanup editing, so the robust construction functions as cheap insurance; its value grows with stronger editors, and the guarantee covers any editor whose effect is dominated by the modeled family.

7 Wave-2 Experiments: Languages, Domains, Paraphrase, Diagnostics

The second wave widens every axis of the validity map and validates the diagnostics of §4. **Scorers (four families):** zero-shot GPT-2 log-likelihood; TF-IDF + logistic regression; a fine-tuned distilroberta-base classifier (trained on the entity-disjoint training half only); and a frozen off-the-shelf detector (Hello-SimpleAI/chatgpt-detector-roberta, used zero-shot). **Data:** all ten MAiDE-up languages (500 human / 300 AI reviews per language; $n_{\text{cal}} = 250$); six text domains (MAiDE-up hotel reviews plus the RAID movie-review, abstracts ($n_{\text{hum}} = 120$), books, news and poetry pools); and an LLM-paraphrase editor (humarin/chatgpt_paraphraser_on_T5_base, a T5-base model fine-tuned on ChatGPT paraphrases). Every condition is evaluated over $R = 200$ calibration/test resamples with the same exact Beta-Binomial violation tests as Wave 1; per-repeat rates are aggregated to means at build time, never hand-entered.

7.1 W1: ten languages — in-language validity is exact; cross-language transfer is bimodal

Table 6: W1: in-language FPR at nominal $\alpha = 0.05$ (four scorers; calibrated and tested within the language) and English-calibrated cross-language FPR.

Language	loglik	tfidf	superv.	off-shelf	X-Eng loglik	X-Eng superv.
Chinese	0.046	0.014	0.049	0.047	0.348	0.000
English	0.049	0.046	0.046	0.049	—	—
French	0.048	0.047	0.048	0.048	0.010	0.000
German	0.049	0.045	0.050	0.047	0.000	0.000
Italian	0.046	0.045	0.046	0.047	0.000	0.000
Korean	0.048	0.006	0.049	0.048	0.996	0.000
Romanian	0.048	0.043	0.049	0.050	0.000	0.000
Russian	0.045	0.042	0.049	0.049	0.998	0.000
Spanish	0.049	0.046	0.049	0.050	0.000	0.000
Turkish	0.047	0.047	0.047	0.052	0.000	0.000

Calibrated within each language, all 40 language–scorer cells sit at or below nominal (FPR 0.006–0.052; no cell rejects the exact test): the certificate’s distribution-freeness is real, and *any* population can be certified by calibrating on it. Deploying the *English*-calibrated certificate cross-lingually is bimodal for the likelihood score: 6 of 9 languages *under*-flag (FPR below nominal — the conservative, power-less direction), while Russian 0.998, Korean 0.996, Chinese 0.348 blow past it, the Cyrillic and Hangul extremes (≈ 1.0) being essentially every human flagged. The mechanism is language identity: GPT-2 assigns non-Latin-script text uniformly low likelihood, which lands either side of the English threshold wholesale. The three classifier-based scores never inflate cross-lingually (max FPR 0.041): they fail safe, at the price of near-zero cross-language power (Figure 8).

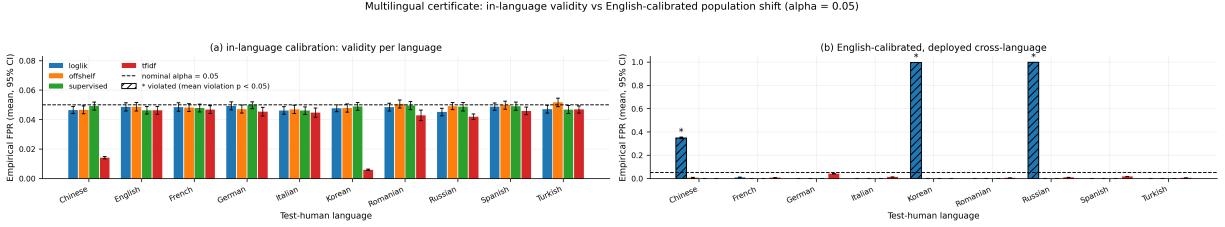


Figure 8: W1: in-language validity (left) vs. English-calibrated cross-language FPR (right) at nominal $\alpha = 0.05$.

7.2 W2: six domains — transfer direction matters

Table 7: W2: in-domain FPR and hotel-calibrated cross-domain FPR at nominal $\alpha = 0.05$.

Target domain	In-domain (loglik)	X loglik	X tfidf	X superv.	X off-shelf
hotel	0.049	—	—	—	—
reviews	0.048	0.433	0.000	0.000	0.050
abstracts	0.046	0.189	0.006	0.000	0.000
books	0.048	0.315	0.000	0.000	0.282
news	0.044	0.855	0.000	0.000	0.169
poetry	0.046	0.061	0.000	0.000	0.018

In-domain calibration is again exact in all six domains. Hotel-calibrated likelihood certificates inflate on edited prose: news 0.85, reviews 0.43, books 0.32, abstracts 0.19 at nominal 0.05, while poetry is mild (0.06) — fluent, professionally edited human text is exactly what a likelihood score mistakes for AI. The supervised scores are conservative under every domain shift (max FPR 0.006), and the off-the-shelf detector sits in between (books 0.28, news 0.17). The full 6×6 source→target matrix (Figure 9) shows the asymmetry: calibrating on the *most* fluent domain (news) is safe everywhere but powerless; calibrating on the least fluent (hotel, poetry) is dangerous everywhere else.

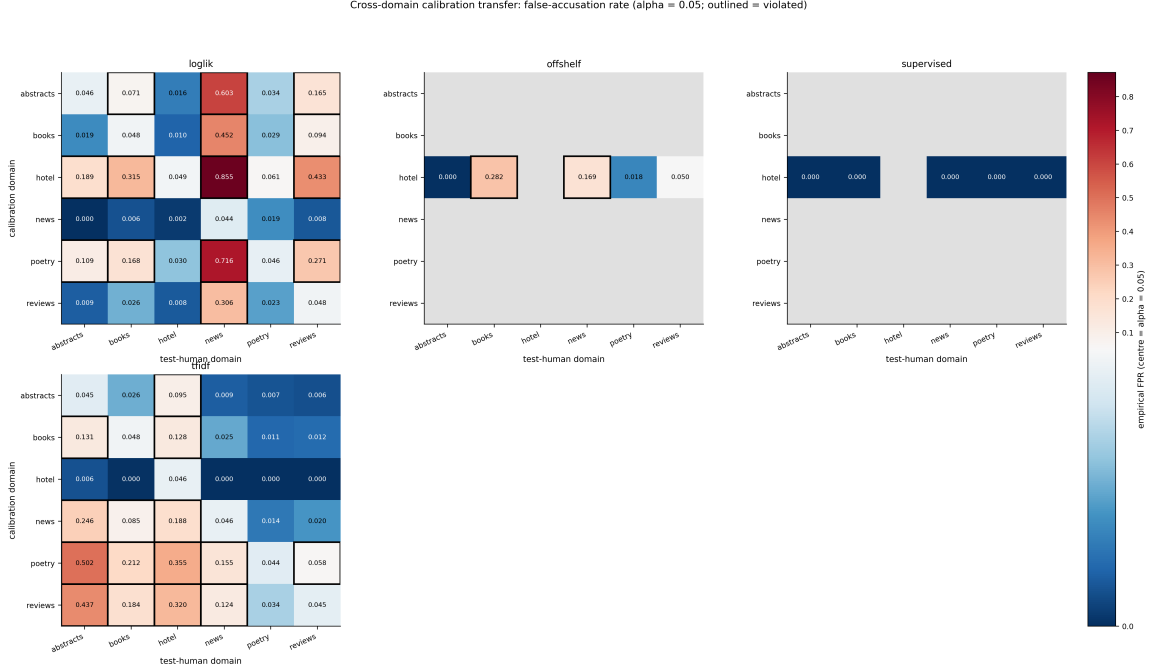


Figure 9: W2: 6×6 cross-domain FPR matrix (calibrate on row, deploy on column) at nominal $\alpha = 0.05$.

7.3 W3: the LLM-paraphrase attack on innocent humans — and its repair

Table 8: W3 at $\alpha = 0.05$: LLM-paraphrased human text under naive vs. paraphrase-aware robust calibration, and the power cost on clean AI text.

Score	FPR clean	FPR para (naive)	FPR para (robust)	TPR (naive)	TPR (robust)
GPT-2 log-likelihood	0.049	0.266	0.024	0.37	0.03
TF-IDF (supervised)	0.051	0.331	0.032	0.88	0.50
DistilRoBERTa (fine-tuned)	0.048	0.554	0.022	0.94	0.36
Off-shelf detector	0.050	0.636	0.023	0.24	0.00

This is the headline failure: a human who innocently runs their own text through an LLM paraphraser (“please improve my wording”) is accused at 0.27–0.64 instead of 0.05 — and unlike the population shifts above, *every* scorer family breaks, including the fine-tuned transformer (0.55) and the off-the-shelf detector (0.64). Paraphrase output *is* LLM output; no scorer can be expected to separate paraphrased-human from generated text, so the fix must come from calibration. Modeling the paraphraser in the editor-aware construction (§3.4; two modeled variants per calibration human, seeds disjoint from test paraphrases) restores validity for all four scorers (0.022–0.032). Unlike the cheap cleanup-editor insurance of E8, here the power cost is substantial — TPR 0.94→0.36 (fine-tuned) and 0.88→0.50 (TF-IDF) on clean AI text — the honest price of remaining unable to accuse paraphrase-shaped humans (Figure 10).

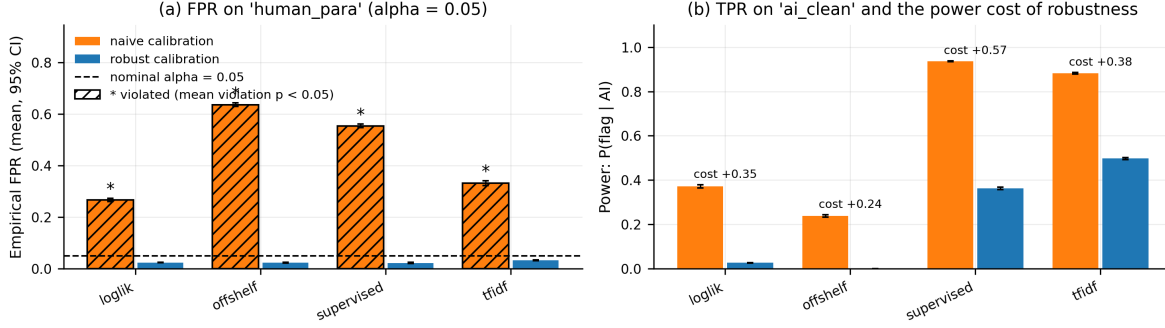


Figure 10: W3: naive vs. paraphrase-aware robust calibration under the LLM-paraphrase attack at nominal $\alpha = 0.05$.

7.4 W4: the diagnostics work — every violation is caught before deployment

Table 9: W4 (log-likelihood score, $\alpha = 0.05$): realized FPR (bold = certified violation), plug-in score-space bounds, and pilot-alarm rate. Alarm level $\rho = 0.1$.

Condition	Realized FPR	TV bound	Γ bound	Alarm@ $m=100$
English control (in-population)	0.048	0.15	0.37	0.06
Chinese humans	0.350	0.55	0.40	1.00
French humans	0.009	0.32	0.28	0.00
German humans	0.000	0.40	0.13	0.00
Italian humans	0.000	0.63	0.27	0.00
Korean humans	0.996	1.02	1.00	1.00
Romanian humans	0.000	0.71	0.50	0.00
Russian humans	0.998	1.03	1.00	1.00
Spanish humans	0.000	0.41	0.13	0.00
Turkish humans	0.000	0.59	0.21	0.00
Reviews humans	0.435	0.66	0.37	1.00
Abstracts humans	0.192	0.60	0.19	0.93 ($m=50$)
Books humans	0.311	0.67	0.28	1.00
News humans	0.852	0.85	0.65	1.00
Poetry humans	0.059	0.26	0.14	0.14
LLM-paraphrased humans	0.263	0.45	0.33	1.00

Across all four scorers, 13 of 64 condition–scorer pairs at $\alpha = 0.05$ are certified violations, and the pilot alarm catches *every one of them*: rate ≥ 0.99 at $m = 100$ unlabeled pilot texts (the small abstracts pool caps its pilot at $m = 50$, where the rate is 0.93). In-control conditions stay at the design level: the English control alarms at most $0.070 \leq \rho = 0.1$, and conservative (under-flagging) shifts never alarm, as they should — the detector tests over-accusation only. The plug-in bounds behave as the theory section warned: the TV bound covers the realized FPR in 15 of 16 likelihood conditions (its one miss is by less than 0.01 on news), while the binned $\hat{\Gamma}\alpha$ bound under-covers in 4 of 16 — plug-in estimates rank risk usefully but must not be read as certificates; the calibrated alarm is the deployable instrument (Figure 11).

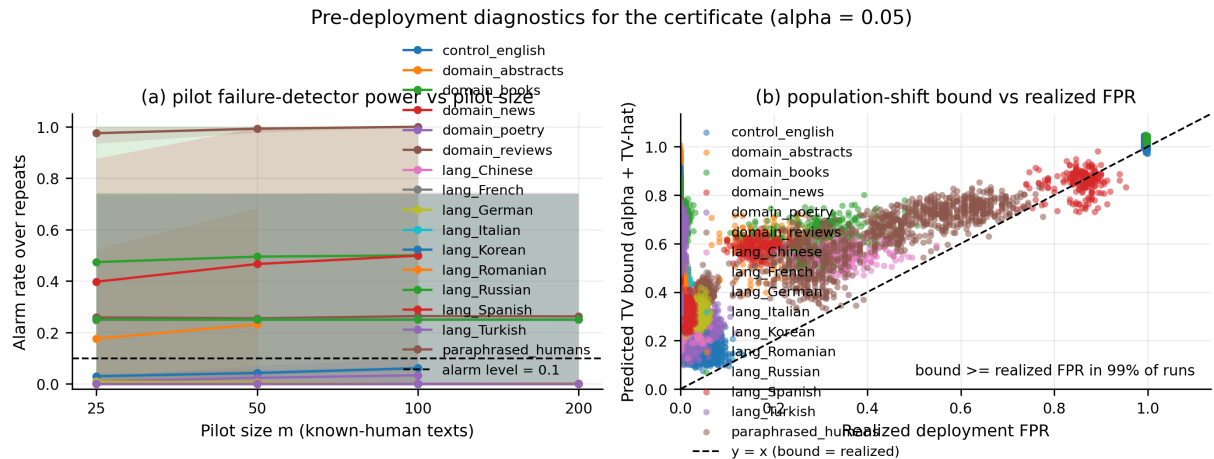


Figure 11: W4: pilot-alarm rate vs. pilot size m per condition (left) and plug-in TV/ $\hat{T}$ bounds vs. realized FPR (right).

8 Discussion

For deployers our results reduce to three rules. First, certify against the humans you will actually judge: the certificate is exact for the calibration population and can be $11\times$ off for a more fluent one; recalibration per population (or per platform) is cheap and necessary. Second, demand conditional validity: marginal certificates can hide systematic over-accusation of long texts (and, by the population result, of fluent writers); Mondrian calibration over observable strata is a one-line fix. Third, choose the score family by failure mode: supervised content scores buy high certified power in-domain and fail safe under shift; zero-shot likelihood scores transfer power across generators but fail dangerous on shifted humans.

9 Limitations

The Wave-1 core is English-only with one in-domain corpus (hotel reviews) whose humans skew non-native — which is what makes the population-shift result vivid, but magnitudes will vary; Wave 2 widens this to ten languages and six domains, and replaces the cleanup editor with a T5 ChatGPT-paraphraser (§7.3), but uses one paraphraser and one multilingual corpus family. GPT-2 is a deliberately modest scorer (the wrapper is scorer-agnostic); our exact tests certify violations, not their causes — the fluency mechanism is supported but not isolated; and the TV/ $\hat{T}$ diagnostics are plug-in estimates whose own sampling error is unquantified here (W4 measures, but does not bound, their coverage). The protocol, released in full, replicates at larger scale directly.

10 Conclusion

Distribution-free certificates make AI-text detectors accountable, but only within the human distribution they were calibrated on. We mapped that boundary — exact in distribution, robust to light editing, broken by population shift, length-biased marginally — and showed the repairs are cheap. Wave 2 stress-tests the map at deployment scale: four scorer families (including a fine-tuned transformer and an off-the-shelf detector), ten languages, six domains, and an LLM-paraphrase attack. The pattern is consistent and honest about its epistemics: validity is *exact in-population by construction* — every language and every domain

certifies cleanly when calibrated on itself — while every cross-population finding is empirical. Likelihood certificates transferred across language or domain fail in both directions (Russian 0.998, news 0.85, versus silent under-flagging elsewhere); LLM-paraphrased innocent humans break *all* scorer families (0.27–0.64 at nominal 0.05) and paraphrase-aware calibration repairs them at a real power cost. Crucially, none of these failures need be discovered in production: the $\Gamma\alpha$ and $\alpha + \text{TV}$ bounds live on the one-dimensional score space, and the calibrated pilot alarm built on the same observation caught every certified violation in this study at $m = 100$ unlabeled pilot texts while respecting its 0.10 false-alarm budget. The bounds remain plug-in estimates, not certificates; the alarm is the deployable instrument. A certificate is a promise about *whom* you calibrated on; deployers should make that promise explicit — and pilot-test it before the first accusation.

Reproducibility. All numbers derive from three JSON files produced by `scripts/run_all.py`, `scripts/run_robust.py` and `scripts/run_wave2.py`; figures and this paper regenerate from them; conformal correctness is asserted by the test suite.

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